

Measuring Hawkish and Dovish Signals in FOMC Communications

A Text-as-Data Analysis of Statements and Minutes, 2000–2024

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Abstract

Central bank communication is a key channel through which monetary policy affects expectations beyond actual interest-rate decisions. This paper studies how the Federal Open Market Committee (FOMC) conveys hawkish and dovish signals in its *Statements* and *Minutes* between 2000 and 2024. We construct a domain-specific dictionary that contrasts terms associated with inflation pressures, tightening, and upside risks with terms related to easing, downside risks, and labor-market weakness. The resulting tone score is computed at both document and sentence level, and is evaluated through several robustness checks, including the masking of explicit rate-decision sentences, dictionary sensitivity tests, TF-IDF classification benchmarks, latent semantic analysis, and Sentence-BERT embeddings. The results show that hawkish language is substantially stronger during rate-hike meetings, while high-volatility periods are associated with a more dovish tone. By contrast, meetings around U.S. elections do not display systematic differences. *Statements* and *Minutes* have similar average tone, but some meetings exhibit large document-level gaps, suggesting that *Minutes* often add nuance to the immediate policy message. Finally, embedding-based representations capture broader semantic and macroeconomic structure, but they do not reproduce the dictionary-based hawkish/dovish axis without supervision. Overall, the results support the use of interpretable, domain-specific lexical measures for tracking monetary-policy tone, while highlighting the limits of generic embeddings for measuring such a targeted economic concept.

1 Introduction

The mandate of the Federal Reserve is dual: it must contain inflation while maintaining a dynamic labor market. To achieve this, it has at its disposal monetary policy tools, notably the management of policy interest rates. Its mission also consists of describing economic conditions, signaling risks, and guiding expectations regarding the future path of monetary policy. This communication channel is particularly visible at the conclusion of meetings of the Federal Open Market Committee (FOMC), when the Committee releases a short Statement, followed several weeks later by more detailed records (*Minutes*). These documents are official, published on a recurring basis, and closely analyzed by financial markets. They therefore constitute a natural corpus for studying monetary policy communication using NLP methods. This article poses a *text-as-data* question: to what extent do FOMC communications convey a “hawkish” or “dovish” signal, and how does this signal vary depending on the type of document (*Statements* / *Minutes*), rate decisions (*Cut* / *Hold* / *Hike*), financial market volatility, electoral periods, and macroeconomic regimes? A “hawkish” signal emphasizes inflation, price pressures, restrictive financial conditions, monetary tightening, or higher interest rates. Conversely, a “dovish” signal highlights support for economic activity, labor market weakness, monetary easing, lower interest rates, or downside macroeconomic risks.

The object of interest (the “hawkish”/“dovish” signal) is latent: it is not directly observed in the data and must be inferred from words, expressions, and the structure of documents. We use a dictionary-based measure as the main estimator and compare it with TF-IDF, LSA, and Sentence-BERT representations. The dictionary score constitutes the central measure, as it corresponds transparently to well-defined economic concepts. TF-IDF classifiers assess

whether information related to rate decisions can also be extracted from a more flexible *bag-of-words* representation. Semantic representations derived from latent semantic analysis (LSA) and Sentence-BERT, in turn, examine whether a dense semantic geometry produces a similar “hawkish/dovish” ranking or instead captures a broader institutional and macroeconomic structure. The results presented in this article are descriptive rather than causal. They show that meetings associated with policy rate increases are clearly more “hawkish” than those with no change or with rate cuts (as expected), and that meetings characterized by a high level of financial volatility, measured by the VIX index, are associated with a less “hawkish” tone (also expected, since volatility is negatively correlated with market returns; thus, periods of crisis or weak growth tend to be associated with higher volatility). Electoral periods do not exhibit a marked average difference. The *Statements* and the *Minutes* display a similar average tone, but differences at the level of individual meetings can be substantial, suggesting that the minutes often contextualize or nuance the immediate message released after the meeting. Finally, combining the “hawkish/dovish” score with an inflation–employment balance indicator shows that the economic diagnosis and the monetary policy signal are closely linked—particularly in the *Minutes*—without constituting the same textual object. The paper makes three contributions. First, it constructs a reproducible processing pipeline that systematically aligns FOMC *Statements* and *Minutes* by meeting and enriches them with financial, political, and macroeconomic context. Second, it proposes a transparent and interpretable measure of monetary policy tone that can be applied at both the document and sentence levels. Third, it compares this manually constructed measure with more flexible text representations derived from machine learning. This comparison separates two tasks that are often conflated: measuring a predefined economic concept and predicting rate-decision labels from text.

2 Related Work

Over the past decades, central bank communication has become an instrument of monetary policy in its own right. Blinder et al. [2008] review the theoretical and empirical evidence and argue that communication can make policy more predictable, influence asset prices, and shape expectations. This literature has helped establish the idea that FOMC texts can be analyzed as economic data rather than as a simple institutional narrative. Several articles use computational methods to study FOMC communication. Lucca and Trebbi [2009] construct automated measures of the content of FOMC statements and show that long-term interest rates respond to communication surprises. This result is particularly relevant for our study, as the same rate decision can be accompanied by different language regarding the future orientation of monetary policy. Hansen et al. [2018] apply NLP methods to FOMC transcripts and show that transparency alters patterns of deliberation and communication. Our article instead focuses on *Statements* and *Minutes*, which are regularly available and comparable over time, but it is grounded in the same perspective that monetary communication can be measured systematically. The article also fits within the broader *text-as-data* literature. Gentzkow et al. [2019] emphasize that textual analysis requires explicit choices regarding representation, measurement, and validation. Loughran and McDonald [2011] show that generic sentiment dictionaries can be misleading in financial corpora, where common words may have domain-specific meanings. For this reason, the main measure proposed here is not an index of positive/negative sentiment, but rather an index of monetary policy tone. The embedding-based checks rely on latent semantic analysis [Deerwester et al., 1990] and on Sentence-BERT [Reimers and Gurevych, 2019], which provide alternative semantic representations of the same documents. This article is methodologically close to dictionary-based approaches, while using machine learning representations as diagnostic tools rather than as the primary estimator. This choice reflects an empirical constraint specific to the corpus. The sample is relatively small by current NLP standards: after alignment, the unit of analysis comprises fewer than two hundred meetings. In the absence of expert-labeled sentences, a fully supervised “hawkish/dovish” classifier would risk learning language specific to particular periods or document formats rather than the underlying monetary policy signal. A dictionary is admittedly less flexible, but it provides a direct and interpretable link between the measured score and the economic vocabulary used to construct it.

3 Data

3.1 FOMC Corpus

The corpus contains FOMC statements and minutes collected from the Federal Reserve website [Board of Governors of the Federal Reserve System, 2024]. Statements are the short announcements released immediately after each meeting. Minutes are longer documents published later; they summarize the discussion, the uncertainty around the outlook, and the arguments behind the decision.

Table 1: Aligned corpus description.

Document type	Meetings	Mean words	Median words	Mean sentences	Date range
Statements	199	526.1	536.0	26.8	2000–2024
Minutes	199	7282.7	7518.0	311.9	2000–2024

Table 2: Rate-decision distribution in the aligned sample.

Decision class	Meetings	Share
Cut	30	15.1%
Hold	110	55.3%
Hike	59	29.6%

The final aligned sample covers 199 meetings from February 2000 to December 2024. Each observation contains a statement, the corresponding minutes, the meeting date and contextual variables. Rate decisions are inferred from the statement text and classified as *hike*, *hold* or *cut*. The sample contains 59 hikes, 110 holds and 30 cuts.

Table 1 shows the large difference in document length. Minutes are roughly fourteen times longer than statements on average. This makes length normalization essential and also explains why statements and minutes should not be expected to carry the same kind of signal. Statements are calibrated policy messages while minutes are richer institutional summaries.

3.2 Alignment and Preprocessing

The raw corpus is organized around the dates of FOMC meetings. Statements and meeting records, i.e. Minutes, are stored in separate fields and then systematically aligned by meeting date. Preprocessing is deliberately conservative in order to preserve as much of the original textual information as possible. Text is converted to lowercase for counting purposes, whitespace is normalized, and recurring fragments originating from web pages (such as navigation menus, update notices, and accessibility statements) are removed when they appear. These steps remove recurring webpage noise while preserving monetary-policy content. The analysis does not remove stopwords that are economically meaningful in the computation of dictionary scores, because several compound expressions depend on local syntax, for example *higher rates*, *downside risks*, and *price stability*. Stopword removal is used only in the TF-IDF benchmark, where the objective is predictive comparison rather than direct term counting. This distinction reflects the conceptual difference between an interpretable dictionary-based measure and a statistical representation designed to maximize classification performance. Documents are also segmented into sentences. Sentence segmentation serves two main functions. First, it allows the masking of sentences that explicitly announce the rate decision, in order to prevent the textual score from mechanically reflecting monetary policy information that is already observed. Second, it enables a dictionary audit at the sentence level by precisely identifying the text segments that contribute to the final score. Sentence-level labels guard against scores driven by a few repeated terms and make the document-level measure easier to audit.

Table 2 reports the class balance. The dominance of hold meetings is expected in monetary policy data, but it matters for evaluation: accuracy alone can be misleading. This is why the predictive experiments also report macro-F1, which gives equal weight to the three decision classes.

3.3 Contextual Variables

In our empirical analysis, we introduce three sets of contextual variables. First, daily data on the VIX volatility index are obtained from Cboe. For each FOMC meeting, the VIX is averaged over a window ranging from five days before to five days after the meeting; the upper quartile of this distribution defines meetings characterized by high financial volatility. This measure is intended to capture episodes of market stress that may influence the tone of monetary communication independently of rate decisions. Second, electoral periods are defined as meetings held within the 90 days preceding a U.S. presidential election or midterm elections. This variable tests whether policy language changes near politically sensitive dates, without assuming strategic behavior.

Third, meetings are also classified according to broad macro-financial regimes corresponding to distinct phases of monetary policy and the macroeconomic environment. These regimes include: (i) the early-2000s easing cycle, marked by the slowdown following the burst of the dot-com bubble; (ii) the global financial crisis of 2008–2009, characterized by systemic financial instability and emergency policy measures; (iii) the zero lower bound (*ZLB*) period, during which policy rates remained close to zero and monetary policy relied more heavily on unconventional instruments; (iv) the normalization phase of 2015–2019, associated with a gradual recovery in economic activity and a progressive tightening of monetary policy; (v) the Covid-19 crisis in 2020, marked by a sharp economic shock and exceptional monetary interventions; (vi) the post-Covid inflation period beginning in 2021, characterized by persistent inflation acceleration and a rapid cycle of monetary tightening; and (vii) a reference category grouping other meetings that do not fall explicitly within these major episodes. These contextual variables are not used to identify causal effects. Rather, they serve to structure the descriptive analysis and to verify whether the textual signal evolves in an economically coherent manner. For example, a meaningful measure of tone should be more “hawkish” during rate hikes than during rate cuts, but it should also reveal more nuanced patterns during periods of stress, when the Federal Reserve may refer to inflationary risks while simultaneously emphasizing financial fragility or downside macroeconomic risks.

4 Methodology and Results

4.1 Dictionary Tone Score

The main model is a dictionary score. Let $H_{d,t}$ and $D_{d,t}$ denote the length-normalized counts of hawkish and dovish terms in document d for meeting t :

$$H_{d,t} = \frac{\#\text{hawkish terms}_{d,t}}{\#\text{words}_{d,t}}, \quad D_{d,t} = \frac{\#\text{dovish terms}_{d,t}}{\#\text{words}_{d,t}}. \quad (1)$$

The normalized tone score is

$$\text{ToneNorm}_{d,t} = \frac{H_{d,t} - D_{d,t}}{H_{d,t} + D_{d,t}}. \quad (2)$$

Positive values indicate a more hawkish document, negative values indicate a more dovish document, and values close to zero indicate a balanced document.

The hawkish dictionary includes terms such as *inflation*, *price stability*, *elevated*, *tightening*, *restrictive*, *higher rates*, *persistent inflation*, and *inflationary pressures*. The dovish dictionary includes terms such as *unemployment*, *weakness*, *slowdown*, *accommodative*, *support*, *lower rates*, *easing*, *recovery*, and *downside risks*. The score is intentionally interpretable: its changes can be audited by reading the terms and the sentences that drive them.

The normalization in Equation (2) is chosen because it is scale-free. A raw difference $H_{d,t} - D_{d,t}$ would still depend on the absolute density of dictionary terms. The normalized balance instead asks whether policy-related language is more hawkish or more dovish conditional on the document containing such language. This is particularly useful when comparing statements and minutes, because minutes contain far more words and far more policy-relevant sentences. When both $H_{d,t}$ and $D_{d,t}$ are zero, the score is set to zero; in practice this is rare for FOMC texts.

4.2 Choice of Model Family

The dictionary score is selected as the primary model because the research question is explicitly oriented toward measurement rather than prediction. The article aims to construct an interpretable time series of monetary policy tone that is comparable across meetings and across document types. For this objective, a transparent and stable measure is preferable to a “black-box” classifier trained on a relatively limited sample. The additional natural language processing models address distinct methodological questions. TF-IDF, combined with linear models, tests whether the text contains sufficient information to predict the observed rate decision, independently of the interpretable structure of the dictionary. Latent semantic analysis (LSA) examines whether documents that are semantically close in a reduced *bag-of-words* space also appear similar along the dimension of monetary policy tone. Sentence-BERT evaluates whether pre-trained contextual embeddings, without task-specific adjustment (*fine-tuning*), spontaneously align with the same “hawkish/dovish” prototype contrast. These approaches are therefore complementary rather than competing. The dictionary is designed to measure an economic concept defined *a priori*, while TF-IDF and embeddings serve as validation and diagnostic tools, making it possible to assess the robustness and coherence of this measure under alternative representations of the corpus.

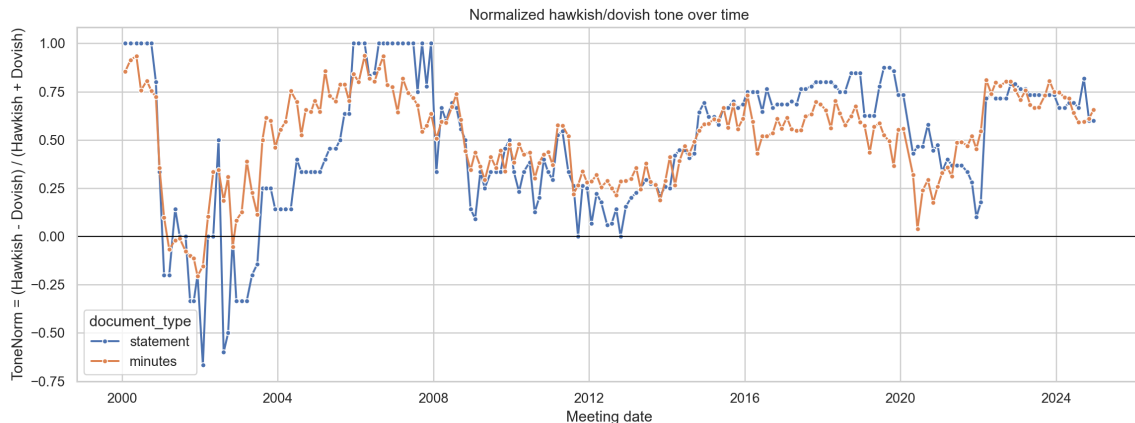


Figure 1: Normalized hawkish/dovish tone in FOMC statements and minutes.

4.3 Validation and Robustness Checks

Four main checks are implemented to assess whether the dictionary captures an economically meaningful monetary policy signal. First, a masked score removes sentences that explicitly announce the rate decision. This procedure makes it possible to test whether the measured tone merely restates the decision—for example, “the Committee raised the target range”—or whether it also appears in the economic assessment and justification surrounding that decision. The objective is to prevent the measure from mechanically reflecting information that is already observed. Second, a sentence-level version assigns each sentence a label—“hawkish,” “dovish,” mixed, or neutral—based on dictionary occurrences. This approach provides both a qualitative analysis of document content and a quantitative measure, at the document level, of the proportion of sentences relevant to monetary policy. It also strengthens the interpretability of the score by explicitly identifying the textual segments that contribute to the measured tone. Third, TF-IDF benchmarks evaluate whether the rate decision can be predicted from the text. Logistic regression and linear SVM models are trained using chronological splits, with earlier meetings used for training and later meetings used for out-of-sample evaluation. This temporal split is more demanding than a random split, because the language of monetary policy evolves over time. Models are estimated on full texts, texts masking the decision, and texts stripped of standardized elements (*boilerplate*), for both *Statements* and *Minutes*. The TF-IDF representation uses unigrams and bigrams, stopword removal, a minimum document frequency of two, and a maximum vocabulary size of 6 000 features. The test set consists of the most recent 30% of meetings, covering the period from June 2017 to December 2024, while the training set covers the period from February 2000 to May 2017. Fourth, dense representations are used as semantic robustness checks. Latent semantic analysis (LSA) is computed by applying truncated singular value decomposition (SVD) to a TF-IDF matrix of document excerpts. A tone score is then defined as the difference between cosine similarity with a “hawkish” prototype and cosine similarity with a “dovish” prototype. Sentence-BERT applies the same prototype logic to contextual embeddings generated by the `all-MiniLM-L6-v2` model. These embedding-based scores serve as robustness checks to determine whether a semantic space less constrained by manual construction produces a comparable ranking of documents. Finally, the study uses a pooled ordinary least squares (*pooled OLS*) regression model as a synthetic descriptive check. The dependent variable is the tone score for each document. Explanatory variables include document type, indicator variables for rate decisions, the standardized VIX, and an electoral period indicator. This model is a descriptive tool intended to verify whether the associations observed in the figures remain present when these different dimensions are taken into account simultaneously.

Figure 1 shows that the average tone is positive in both document types. The mean normalized score is 0.484 for statements and 0.507 for minutes. This means that, over the full sample, language related to inflation, price pressure and restrictive conditions is more frequent than language related to easing or weakness. The score falls in the early 2000s and in crisis periods, then rises during the 2015–2019 normalization and the post-Covid inflation episode.

The masked score is very close to the main score: the means are 0.474 for statements and 0.501 for minutes. The policy signal is therefore not driven only by direct decision sentences. The same gradient appears at sentence level: policy-relevant sentences account for about 31.0% of statements sentences and 29.7% of minutes sentences, and their

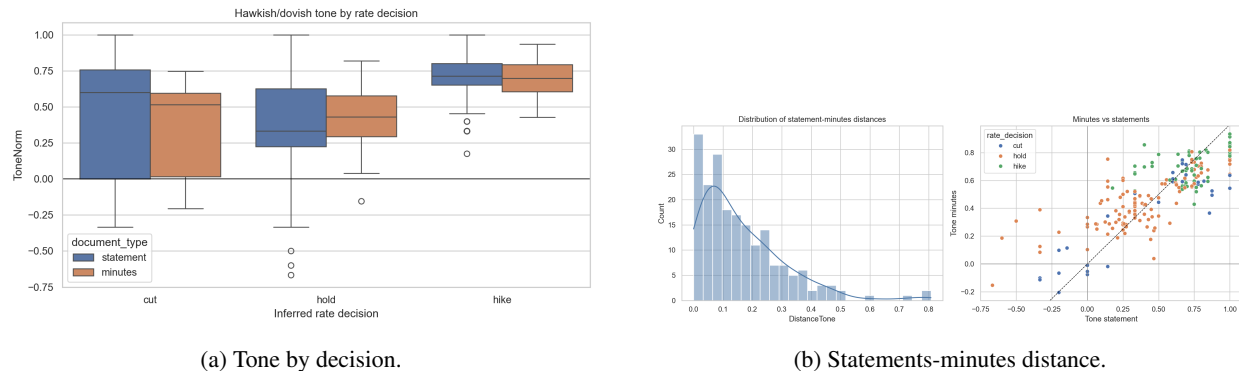


Figure 2: Document tone by rate decision and meeting-level distance between statements and minutes.

Table 3: Average normalized tone by rate decision.

Document type	Cut	Hold	Hike
Statements	0.418	0.384	0.705
Minutes	0.368	0.442	0.700

balance is highest during hike meetings.

The time series shows why average tone alone is insufficient. During the zero lower bound period, policy was highly accommodative, yet the documents continued to discuss price stability and inflation expectations. During the post-Covid inflation period, the language becomes more directly hawkish because the inflation diagnosis and the tightening signal point in the same direction. The dictionary therefore behaves less like a generic sentiment index and more like a stance indicator: it reacts to the mixture of inflation-risk language and accommodation language, not to whether the economic outlook is described as “good” or “bad”.

4.4 Statements, Minutes and Rate Decisions

Hike meetings are clearly more hawkish than hold and cut meetings. Table 3 reports average scores. For statements, hike meetings have an average tone of 0.705, compared with 0.384 for holds and 0.418 for cuts. For minutes, the corresponding values are 0.700, 0.442 and 0.368. The lower average score for cut minutes is economically intuitive: minutes written around easing episodes emphasize weakness, downside risks and support more than hike minutes do. The statements-minutes distance is

$$\text{DistanceTone}_t = |\text{ToneNorm}_{\text{minutes},t} - \text{ToneNorm}_{\text{statements},t}|. \quad (3)$$

Its mean is 0.163 and its median is 0.122. A paired comparison does not indicate a large systematic difference in average tone between statements and minutes, but the meeting-level distribution is informative. Several meetings, especially in the early-2000s easing cycle, display large gaps. In those cases, the Statement can be strongly dovish while the minutes remain more balanced because they record competing risks or a more detailed debate.

Statements and Minutes differ not only in length but also in function. Statements are written as coordinated public signals. They must summarize the Committee’s decision in a compact and market-readable form. Minutes, by contrast, are designed to document the reasoning behind the decision. They can contain hawkish and dovish arguments in the same document, especially when the Committee faces a trade-off between inflation concerns and real-activity weakness. A large statements-minutes distance should therefore not be interpreted automatically as inconsistency. It is better read as evidence that the later document adds nuance to the immediate signal.

4.5 Financial and Political Context

High-volatility meetings are associated with lower tone scores. In meetings below the upper VIX quartile, the average score is 0.542 for statements and 0.549 for minutes. In high-VIX meetings, the averages fall to 0.311 and 0.382. This

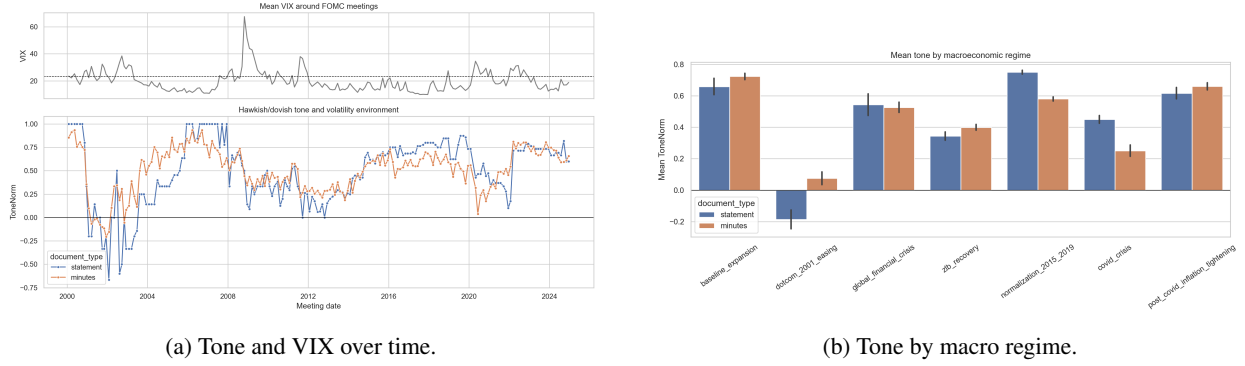


Figure 3: Tone across financial stress and macro-financial regimes.

Table 4: Descriptive OLS model for normalized tone. The reference category is a statement during a hold meeting.

Variable	Coefficient	<i>p</i> -value
Intercept	0.406	0.000
Minutes	0.023	0.376
Hike decision	0.255	0.000
Cut decision	0.012	0.755
VIX, standardized	-0.082	0.000
Election window	0.002	0.945

pattern is consistent with the Fed using more dovish or support-oriented language during episodes of financial stress. In contrast, election windows show little average difference: statement tone is 0.494 outside election windows and 0.459 inside them; tone in Minutes is 0.511 outside and 0.497 inside.

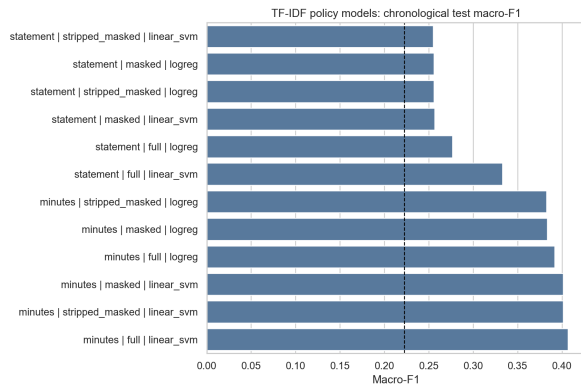
The macro-regime analysis is consistent with the time series. The normalization period and the post-Covid inflation period are among the most hawkish regimes. The Covid period and the early-2000s easing cycle are less hawkish. The global financial crisis is not the most dovish regime in the dictionary score because those documents also contain frequent references to inflation, price stability and financial strains. The global financial crisis shows that weakness language and inflation-risk language can coexist in the same document.

The absence of a strong election-window pattern is also informative. In this descriptive design, the tone series does not suggest an obvious shift in the language of policy around presidential or midterm elections. This should not be overinterpreted as evidence that politics never affects communication. The election variable is deliberately simple, and the number of meetings inside election windows is limited. The result is narrower: conditional on this corpus and this dictionary score, the most visible variation is linked to policy and financial conditions, not to the election-window indicator.

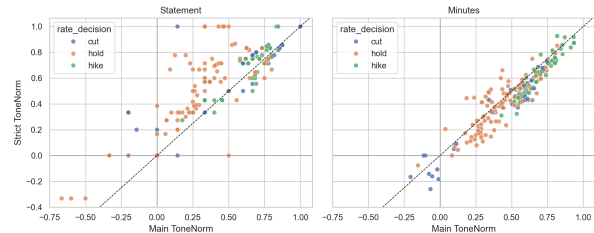
4.6 Regression Evidence

Table 4 summarizes the descriptive associations in a pooled document panel. The hike coefficient is large and statistically significant, which confirms that the tone score is aligned with the most tightening-oriented decision class. The VIX coefficient is negative and significant. The minutes dummy is positive but not statistically significant, consistent with the paired comparison. The election coefficient is essentially zero in this specification.

The cut coefficient is small in the pooled specification, even though cut minutes are visibly less hawkish in the grouped averages. The small cut coefficient reflects heterogeneity across easing episodes. Some cuts occur in crisis periods dominated by support and downside-risk language; others occur when inflation concerns have not fully disappeared. The tone score therefore captures the communication surrounding the decision, not only the sign of the rate change.



(a) TF-IDF chronological test macro-F1.



(b) Main and strict dictionary scores.

Figure 4: Representation and dictionary robustness checks.

Table 5: Summary of representation checks.

Check	Statements result	Minutes result
Masked mean tone	0.474	0.501
Strict dictionary correlation with main score	0.839	0.935
Best TF-IDF chronological macro-F1	0.333	0.407
LSA prototype correlation with dictionary tone	0.407	0.150
Sentence-BERT prototype correlation with dictionary tone	0.043	0.048

4.7 NLP Model Checks

The TF-IDF models provide a benchmark for whether rate-decision information is present in the text beyond the handcrafted score. The best chronological test performance is obtained with minutes and a linear SVM: 0.550 accuracy and 0.407 macro-F1. The same model on masked minutes reaches the same accuracy and 0.401 macro-F1, which again suggests that the decision signal is not confined to the explicit announcement sentence. Models trained on Statements perform worse, with the best macro-F1 around 0.333. This gap is plausible because statements are short and formulaic, while minutes contain more evidence about the economic reasoning behind the decision.

The TF-IDF coefficients are useful for auditing the model. Hike-associated terms include inflation, prices, energy prices and point increases. Cut-associated terms include discount-rate reductions and crisis-era action vocabulary. Hold-associated terms include purchases, recovery, accommodative stance and asset programs. The terms are economically interpretable, but they also reveal a limitation: a flexible text classifier can partly learn episode-specific vocabulary, not only policy stance.

Dictionary sensitivity is checked by removing broad words such as generic *increase*, *pressure* and *support*. The main and strict scores remain strongly related. The correlation is 0.839 for statements and 0.935 for minutes. The mean absolute difference is larger for statements (0.159) than for minutes (0.079), which is expected because short documents are more sensitive to small dictionary changes. The ordering by rate decision remains stable.

Table 5 summarizes the main robustness checks. The strict dictionary result supports the stability of the lexical score. The TF-IDF result shows that minutes contain more predictive information about the policy decision than statements do. The embedding correlations are weaker, especially for minutes, which suggests that dense semantic structure is not equivalent to the explicit hawkish/dovish concept measured by the dictionary.

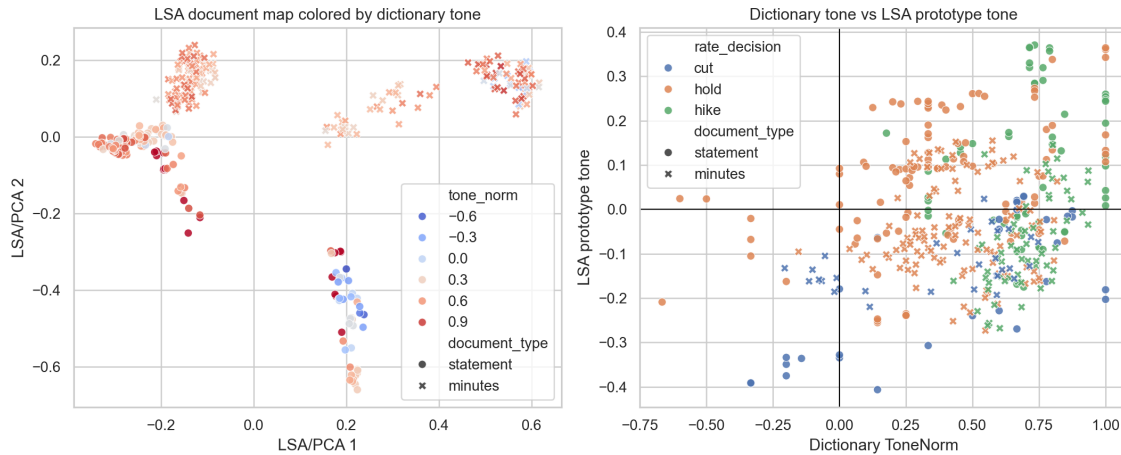


Figure 5: LSA document geometry and prototype-based tone.

4.8 Embedding Robustness

Figure 5 compares the dictionary score with an LSA prototype score. The first 30 LSA components explain 51.5% of TF-IDF variance. The correlation between LSA prototype tone and dictionary tone is 0.407 for statements and 0.150 for minutes. This positive but moderate relationship suggests that LSA captures some policy-tone information, especially in short statements, while minutes’ dense geometry is shaped by broader topics and document structure.

Nearest-neighbor results lead to a similar interpretation. For selected meetings, neighboring documents often come from similar policy episodes, such as crisis meetings, zero lower bound communications, or inflation-tightening periods. This is useful for qualitative exploration: embeddings can retrieve comparable communication episodes without relying on the dictionary. However, the neighbors are not always ordered by hawkishness. They may instead be close because they share institutional vocabulary, document format or crisis-specific topics. For this reason, embeddings are kept as robustness checks rather than used as the main measure.

The Sentence-BERT prototype check produces very weak correlations with the dictionary tone: 0.043 for statements and 0.048 for minutes. This means that a generic contextual embedding, combined with short hawkish and dovish prototypes, does not reproduce the explicit monetary-policy lexicon. In this application, Sentence-BERT is better viewed as a tool for qualitative nearest-neighbor inspection or clustering of communication episodes than as a direct substitute for the interpretable dictionary score.

Sentence-BERT embeddings are trained to encode broad sentence similarity, not to estimate central-bank stance. Without fine-tuning or labelled examples, the model has no reason to place *price stability*, *restrictive stance*, and *downside risks* on the exact monetary-policy axis needed here. The weak prototype correlations therefore support the choice of a domain-specific measure. A supervised embedding model could be a natural extension, but it would require expert labels at the sentence or paragraph level.

4.9 From Economic Diagnosis to Policy Signal

The final analysis relates the hawkish/dovish score to a separate inflation/employment balance. The correlation between mandate balance and tone is small for statements (-0.042) but positive for minutes (0.405). The relationship between diagnosis and tone is therefore much stronger in Minutes than in Statements. Statements are concise and strongly calibrated around the immediate policy message, so the diagnostic balance and the policy signal can diverge. Minutes provide more space for connecting inflation, labor-market conditions and the policy stance; the two dimensions therefore move together more clearly.

Most meetings fall in the inflation-hawkish quadrant. This reflects both the importance of price stability in FOMC communication and the design of the hawkish/dovish dictionary. More interesting are the off-diagonal cases: meetings where inflation language is salient but the tone is relatively dovish, or where employment concerns coexist with hawkish policy language. These cases are useful for qualitative interpretation because they identify moments when the diagnosis and the policy signal do not point in the same direction.

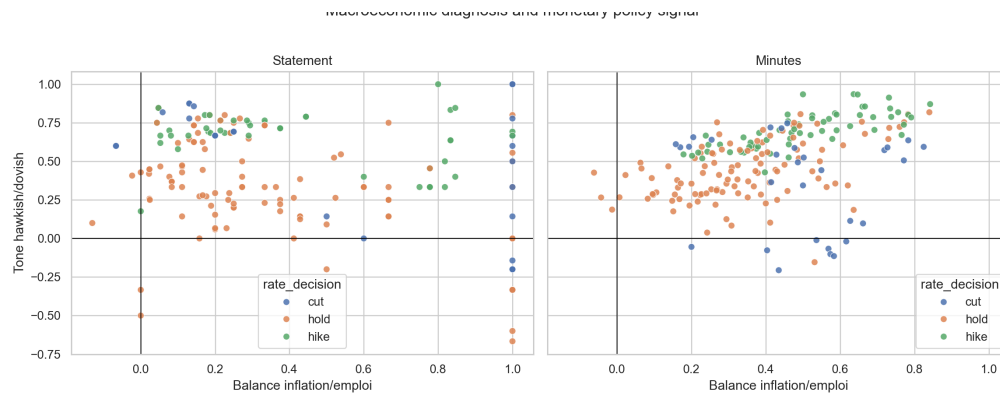


Figure 6: Inflation/employment balance and hawkish/dovish tone.

The two-dimensional view separates the Fed’s diagnosis from its policy stance. A document can be inflation-focused because the Committee is explaining high realized inflation, but the policy signal can still be cautious if financial conditions are fragile. The opposite case can occur when labor-market language remains prominent while the Committee signals that policy must stay restrictive. The scatter plot therefore turns a one-dimensional tone series into a map of communication regimes.

5 Discussion of results

The dictionary score provides a transparent and economically interpretable measure of monetary-policy tone. It reproduces the expected ordering of rate decisions, remains robust to the masking of sentences announcing the decision, and remains stable under a stricter version of the dictionary. It also makes it possible to compute interpretable distances, at the meeting level, between *Statements* and *Minutes*. The results also shed light on the role of document type. The *Statement* is the high-visibility message released immediately after a meeting. It condenses the Committee’s decision and assessment into a short, carefully negotiated text. The *Minutes*, by contrast, constitute a delayed explanation of the discussion. The fact that their average tone is not statistically different from that of the statements should not obscure their analytical value. Their length makes them more useful for identifying the underlying reasons behind the signal, and their meeting-level distance from the statements helps identify situations in which the public message was subsequently qualified or refined through more detailed arguments. Several limitations remain. Dictionary-based scores depend on the selected terms, and some words are sensitive to context. A term such as “support” may describe an accommodative policy, but it may also appear in a neutral institutional expression. Conversely, a sentence may be “hawkish” without explicitly using any dictionary term. Checks based on a stricter version of the dictionary and on sentence-level analysis reduce this concern, but they do not eliminate it entirely. Another limitation concerns the unit of observation. The analysis treats each document as a single observation, whereas meeting records contain multiple speakers, themes, and sections. A more granular model could distinguish staff analyses, participants’ views, and the monetary policy discussion itself. It could also differentiate uncertainty language from policy-stance language. This distinction would be particularly relevant during meetings in which the Federal Reserve simultaneously describes weak economic activity, high uncertainty, and inflationary risks. Supervised checks and embedding-based analyses clarify what the dictionary measures—and what it does not measure. TF-IDF classifiers confirm that information related to monetary policy decisions is indeed present in the text, particularly in the meeting records, but they also learn vocabulary specific to certain historical episodes. LSA and Sentence-BERT capture semantic neighborhoods and macroeconomic episodes, but they do not spontaneously reproduce the “hawkish/dovish” scale without stronger supervision. As an extension, we could manually annotate sentences as “hawkish,” “dovish,” or neutral, and then train a supervised classifier on expert labels. Another extension would be to analyze financial market reactions around meetings in order to distinguish descriptive tone from genuine communication surprises. The approach may transfer to other central banks, but only after adapting the dictionary to each institution. The same processing pipeline could be applied to European Central Bank press conferences or Bank of England meeting records, but the dictionaries would need to be adapted to the institutional language and monetary policy frameworks specific to each institution. Even within the Federal Reserve corpus, vocabulary evolves over time. Terms related to asset purchases, forward guidance,

and balance sheet management become central after 2008. A static dictionary remains easy to interpret, but future work could allow the vocabulary to evolve in a controlled manner while preserving a verifiable correspondence with the “hawkish” and “dovish” concepts.

6 Conclusion

This paper measures the hawkish/dovish signal in FOMC statements and minutes from 2000 to 2024. Hawkish language peaks around rate hikes, declines in high-VIX periods, and shows little variation around election windows. Minutes and statements have similar average tone but can differ substantially at the meeting level. Robustness checks based on decision masking, sentence tagging, TF-IDF models, strict dictionaries, LSA and Sentence-BERT support the central interpretation: the dictionary score is a useful, transparent measure of monetary-policy signal, while dense embeddings are better suited to studying broader semantic structure and communication regimes.

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